

JEE-Main-27-01-2024 (Memory Based)

[EVENING SHIFT]

Physics

Question: Find Dimensions of $\left[\frac{a}{b^2} \right] \left(P + \frac{a}{V^2} \right) (V - b) = nRT$

Options:

- (a) $[M^1 L^{-1} T^{-2}]$
- (b) $[M^2 L^2 T^{-2}]$
- (c) $[M^1 L^1 T^{-2}]$
- (d) $[M^0 L^2 T^{-2}]$

Answer: (a)

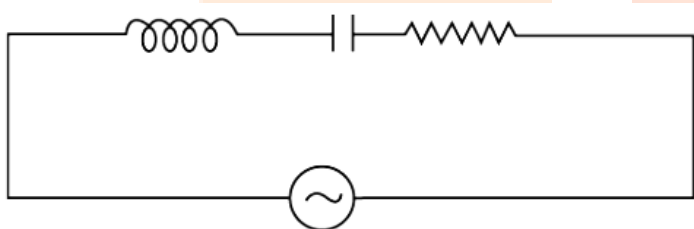
Question: Kinetic energy of 1 mole of oxygen is?

Options:

- (a) 3735
- (b) 6225
- (c) 1245
- (d) 2000

Answer: (b)

Question: An LCR series AC circuit with $L = \frac{100}{\pi} \text{ mH}$, $C = \frac{10^{-3}}{\pi} \text{ F}$ and $R = 10\Omega$ has $\omega = 50$ Hertz. The voltage has RMS value of 230 volts. Find the power Factor of the circuit

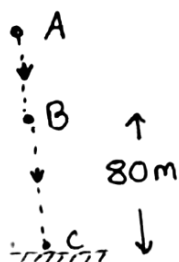


Options:

- (a) 0
- (b) 1
- (c) $\frac{1}{2}$
- (d) $\frac{\sqrt{3}}{2}$

Answer: (b)

Question: An object is released from point A time from B to C is 2s . Find height AC?



Options:

- (a) 45 m
- (b) 100 m
- (c) 125 m
- (d) 160 m

Answer: (c)

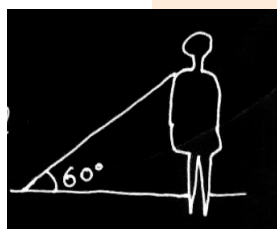
Question: Galvanometer shows deflection of $\pi/3$ where $I_1 = 200 \mu\text{A}$ current is passed what will be the current when the deflection in galvanometer is $\pi/10$

Options:

- (a) $600 \mu\text{A}$
- (b) $60 \mu\text{A}$
- (c) $20 \mu\text{A}$
- (d) $40 \mu\text{A}$

Answer: (b)

Question: M rod = 12 kg. Find normal by shoulder? Ground is rough

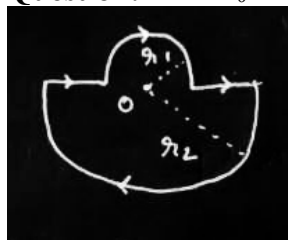


Options:

- (a) 30N
- (b) 60N
- (c) 15N
- (d) 120N

Answer: (a)

Question: Find B_0 ? Current = 4A, $r_2 = 4\pi \text{ m}$, $r_1 = 2\pi \text{ m}$

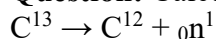


Options:

- (a) 2×10^{-7}
- (b) 10^{-7}
- (c) 4×10^{-7}
- (d) 3×10^{-7}

Answer: (a)

Question: Calculate the Q-value of the following nuclear reactions in Mega electron volts



Use the following data for masses

$${}^{13}\text{C} = 13.009 \text{ u}$$

$${}^{12}\text{C} = 12 \text{ u}$$

$$\text{n} = 1.008 \text{ u}$$

Options:

(a) 0.93 MeV

(b) 1.93 MeV

(c) 2.93 MeV

(d) 3.93 MeV

Answer: (a)

Question: S1 : Moon takes less time to revolve around the earth as compared to earth around the sun

S2: Angular velocity of moon w.r. to earth is greater than angular velocity of earth w.r to sun

Options:

(a) Both are true and S2 is the correct explanation of S1

(b) Both are true and S2 is not the correct explanation of S1

(c) S1 is true, S2 is false

(d) S1 is false, S2 is true

Answer: (a)

Question: Refractive index of a medium is 3 and speed of light in air is 3×10^8 m/s. Intensity of a beam incident on a non reflecting surface ($r = 0$) kept in the medium is I. Find the pressure on the surface. (consider normal incidence)

Options:

(a) $I/10^8 \text{ N/m}^2$

(b) $2I/10^8 \text{ N/m}^2$

(c) $3I/10^8 \text{ N/m}^2$

(d) $4I/10^8 \text{ N/m}^2$

Answer: (a)

Question: $\phi = 6.63 \text{ eV}$ find threshold frequency?

Options:

(a) $1.6 \times 10^{15} \text{ Hz}$

(b) $1.6 \times 10^{16} \text{ Hz}$

(c) $3.2 \times 10^{15} \text{ Hz}$

(d) $1.6 \times 10^{14} \text{ Hz}$

Answer: (a)

Question: A bullet loses to $\frac{1}{3}$ velocity while travelling 4 cm in a wooden block. it travels a further distance of $D \times 10^{-3} \text{ m}$ and stops find D.

Options:

(a) 1

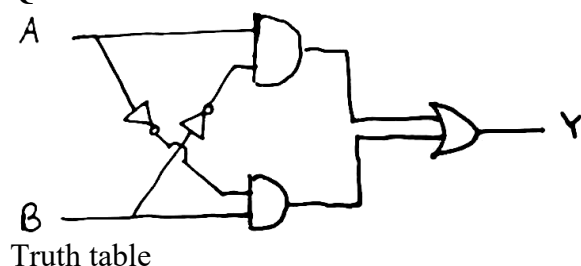
(b) 8

(c) 5

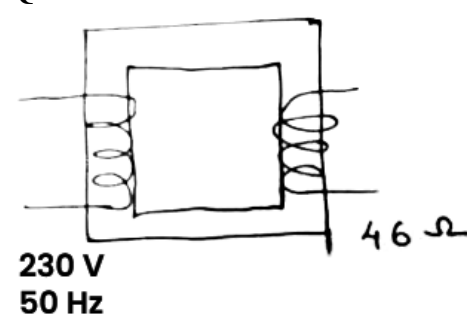
(d) 6

Answer: (5)

Question:



Question:



Turns ratio 10 : 1. Find power supplied.

Question: An Electron in Hydrogen atom emits a photon of Paschen Series with maximum possible energy. What is the wavelength of the photon?

Options:

- (a) 821 nm
- (b) 624 nm
- (c) 1023 nm
- (d) 121 nm

Answer: (a)

Question: Closed organ pipe of length 1.5 m and an open organ pipe of length 3.5 m produce a beat frequency of 7 in their fundamental mode. Find the speed of sound.

Options:

- (a) 330 m/s
- (b) 294 m/s
- (c) 404 m/s
- (d) 350 m/s

Answer: (b)

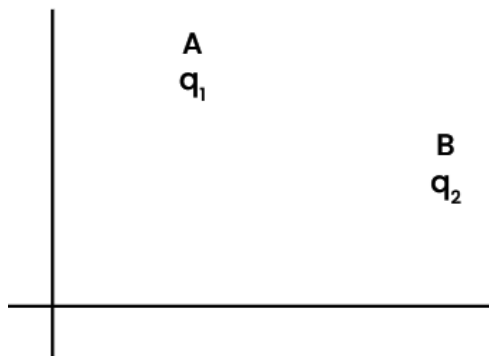
Question: Two charges of $q_1 = -4\mu\text{C}$ and $q_2 = 4\mu\text{C}$ are placed at the locations shown, in an electric field of

$\vec{E} = 0.2\text{v} / m\hat{i}$. Find the torque experienced by the dipole

Coordinate of A (1, 0, 3)

Coordinate of B = (2, 0, 4)

Electric field $\vec{E} = 2\hat{i}$

**Options:**

- (a) $8 \times 10^{-6} \text{ N-m}$
- (b) $4 \times 10^{-6} \text{ N-m}$
- (c) 10^{-6} N-m
- (d) $2 \times 10^{-6} \text{ N-M}$

Answer: (a)**Question:** Read the following assertion and reason and then choose the correct option.

Assertion: If a body regains its original shape and size after deforming force is removed it is said to be a plastic body

Reason: The restoring force of a body depends on the interatomic distances

Options:

- (a) Both Assertion and Reason are correct and reason explains assertion
- (b) Both Assertion and Reason are correct but reason does not explain assertion
- (c) Only Assertion is correct
- (d) Only Reason is correct

Answer: (d)**Question:** Read the following statement and then choose the correct option:

S1: Limiting value of static friction is dependent on area of contact and independent of nature of material

S2: Limiting value of kinetic friction is independent of area of contact and dependent on nature of material

Options:

- (a) Both S1 and S2 are correct
- (b) Both S1 and S2 are wrong
- (c) Only S1 is correct
- (d) Only S2 is correct

Answer: (d)**Question:** An adiabatic process a gas follows the process

$P \propto T^3$. Find its adiabatic exponent $\left(r = \frac{C_p}{C_v} \right)$

Options:

- (a) 1.5
- (b) 1.33
- (c) 1.4
- (d) 1.66

Answer: (a)

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[EVENING SHIFT]

Chemistry

Question: Which of the following cannot act as an oxidising agent?

Options:

- (a) MnO_4^-
- (b) SO_4^{2-}
- (c) N_3^-
- (d) BrO_3^-

Answer: (c)

Solution: N_3^-

Question: The quantity which changes with temperature:

Options:

- (a) Mole fraction
- (b) Mass Percentage
- (c) Molarity
- (d) Molality

Answer: (c)

Solution: Molarity is dependent on temperature. As the temperature increases, water expands, so the solution's volume therefore increases

Question: Phenolic group can be identified by a positive

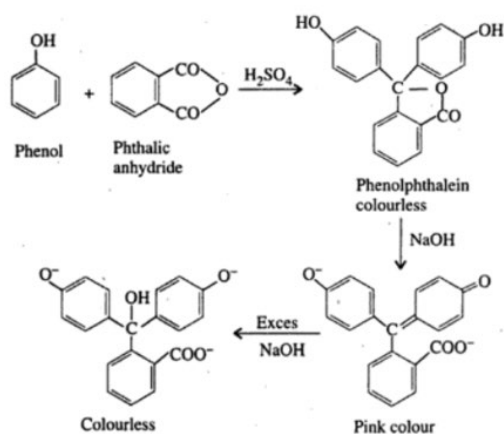
Options:

- (a) Lucas test
- (b) Carbylamine test
- (c) Phthalein test
- (d) Tollen's test

Answer: (c)

Solution:

Phthalein Dye Test



Question: Find the longest wavelength in Paschen series terms of R

Options:

- (a) $144/7R$
- (b) $123/2R$
- (c) $170/R$
- (d) $16/R$

Answer: (a)

Solution:

$$\frac{1}{\lambda} = R \left(\frac{1}{3^2} - \frac{1}{4^2} \right) = R \frac{7}{144}$$

Question: First order reaction 99.9 % completion and half life ratio?

Options:

- (a) 10
- (b) 5
- (c) 20
- (d) 4

Answer: (a)

Solution:

$$t = \frac{2.303}{k} \log \frac{a}{a-x}$$

$$(i) \quad t_1 = \frac{2.303}{k} \log \frac{100}{100-99.9} \text{ (for 99.9% completion)}$$

$$= \frac{2.303}{k} \log \frac{100}{0.1}$$

$$= \frac{2.303}{k} \times 3$$

$$(ii) \quad t_2 = \frac{2.303}{k} \log \frac{100}{100-50} \text{ (for 50% completion)}$$

$$= \frac{2.303}{k} \log 2$$

$$\frac{t_1}{t_2} = \frac{3}{0.3010} \approx 10$$

Question: S1: Ce^{4+} is stable because of noble gas configuration

S2: Ce^{4+} is good R. A. as it can go to +3 O.S.

Options:

- (a) Statement I is incorrect but statement II is correct
- (b) Both statement I and II are correct
- (c) Both statement I and II are incorrect
- (d) Statement I is correct but statement II is incorrect

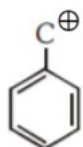
Answer: (d)

Solution: Statement I is correct but statement II is incorrect

Question: Which of the following not undergo S_N1 ?

Options:

- (a) $C = C^{\oplus}$
- (b) $2^\circ C^{\oplus}$
- (c)



(d) $C = C - C^{\oplus}$

Answer: (a)

Question: C_2H_6 newman projection find incorrect information

Options:

- (a) Infinite conformers
- (b) Interconvertible
- (c) Dihedral angle in staggered 60°
- (d) Eclipsed is more stable.

Answer: (d)

Solution: Staggered form is more stable

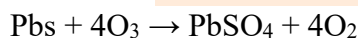
Question: 1 mole of Pbs reacts with x mol of O_3 to give y moles of O_2 then $x + y$?

Options:

- (a) 8
- (b) 9
- (c) 4
- (d) 6

Answer: (a)

Solution:



$$x = 4 \quad y = 4$$

$$x + y = 8$$

Question: Which structure of protein is intact after coagulation of egg white on boiling?

Options:

- (a) Primary
- (b) Secondary
- (c) Tertiary
- (d) Quaternary

Answer: (a)

Solution: Denaturation of protein causes structural change in secondary & tertiary structure of protein but primary structure remain unchanged.

Question: The molecular formula of second homologue in the homologous series of monocarboxylic acid is

Options:

- (a) CH_3COOH
- (b) CH_3CH_2COOH
- (c) $CH_3CH(CH_3)COOH$
- (d) $CH_3CH_2CH_2COOH$

Answer: (a)

Solution: 1st homologue $HCOOH$

2nd homologue CH_3COOH

Question: The technique used for purification of steam volatile water immiscible substance is:

Options:

- (a) Fractional Distillation
- (b) Distillation under reduced pressure
- (c) Steam Distillation
- (d) Simple Distillation

Answer: (c)

Solution: Steam distillation method is used to separate substances which are steam volatile and are immiscible with water. However, the impurities should not be steam volatile in order to purify the substance by steam distillation.

Question: In which of the options all the elements have d^{10} configuration in their ground state

Options:

- (a) Cu, Zn, Cd, Ag
- (b) Cd, Au, Hg, Ni
- (c) Sc, Ti, Fe, Zn
- (d) Fe, Cr, Co, Ni

Answer: (a)

Solution:

Zn - $3d^{10} 4s^2$

Cu - $3d^{10} 4s^1$

Cd - $4d^{10} 5s^2$

Ag - $4d^{10} 5s^1$

Question: Number of non - polar molecules

H_2O , CH_4 , SO_2 , $CHCl_3$, PF_3 , NH_3 , SO_2 , HF

Answer: 2

Solution: Number of non - polar molecules = 2

CH_4 , CO_2

Question: How many of them have d^2sp^3 Hybridisation?

- (a) $[Co(NH_3)_6]^{3+}$
- (b) $[PtCl_6]^{2-}$
- (c) SF_6
- (d) BrF_2^-

Answer: 2

Solution:

(a) $[Co(NH_3)_6]^{3+}$

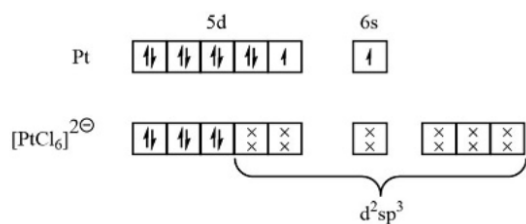
(b) $[PtCl_6]^{2-}$

$Co^{3+} - 3d^6$

In presence of NH_3 ligand pairing of electron takes place

$[Co(NH_3)_6]^{3+}$ have d^2sp^3 Hybridisation

$Pt^{4+} - 5d^6$



Question: How many have noble gas configuration?

- (a) Fe^{2+}
 (b) Cs^+
 (c) Sr^{2+}
 (d) Pb^{2+}

Answer: 2

Solution:

$\text{Cs}^+ - [\text{Xe}]$

$\text{Sr}^{2+} - [\text{Kr}]$

Both have noble gas configuration.

Question: In a standard Hydrogen Electrode, $\text{pH} = 3$

What is the EMF of the electrode in this case?

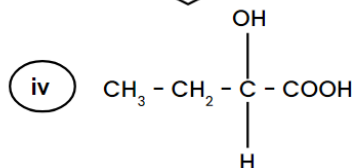
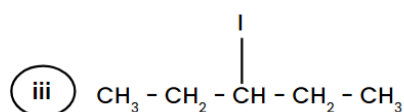
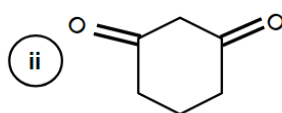
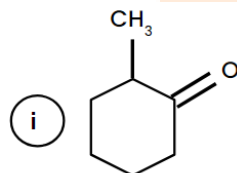
Solution:

$E_{\text{H}^+ / \text{H}_2} = -0.0591 \text{ pH} \quad \because -\log [\text{H}^+] = \text{pH}$

$$= -0.0591 \times 3$$

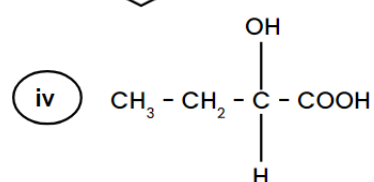
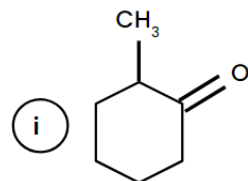
$$= -0.1773$$

Question: How many compound(s) given below have chiral carbon ?



Answer: 2

Solution:



Question: What volume of 3M NaOH solution can be formed using 84g of NaOH

Solution:

$$\text{Moles of NaOH} = \frac{84}{40} = \frac{21}{10} = 2.1$$

$$M \times V = 2.1$$

$$3 \times V = 2.1$$

$$V = 0.7 \text{ ltr} = 700 \text{ ml}$$



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[EVENING SHIFT]

Mathematics

Question: If 20th term from the end of progression $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \dots, 129\frac{1}{4}$ is ____.

Options:

- (a) -120
- (b) -115
- (c) -125
- (d) -110

Answer: (b)

Solution:

$$T_{20} = a + (n-1)d$$

$$= -129\frac{1}{4} + 19 \times \frac{3}{4}$$

$$= -115$$

Question: $P = (1-x)^{2008} (1+x+x^2)^{2007}$; find the coefficient of x^{2012} ?

Answer: 0.00

Solution:

$$(1-x)^{2008} (1+x+x^2)^{2007}$$

$$= (1-x)(1-x^3)^{2007}$$

$$= (1-x) \sum_{r=0}^{2007} {}^{2007}C_r (-x^3)^{2007-r}$$

$$\text{Coefficient of } x^{2012} = 0$$

Question: The integral $\int \frac{(x^8 - x^2)}{(x^{12} + 3x^6 + 1) \tan^{-1}\left(x^3 + \frac{1}{x^3}\right)} dx$ is equal to:

Options:

(a) $\frac{1}{3} \ln \left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| + C$

(b) $\ln \left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| + C$

$$(c) \frac{1}{6} \ln \left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| + C$$

$$(d) \frac{1}{9} \ln \left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| + C$$

Answer: (a)

Solution:

$$\begin{aligned} & \int \frac{(x^8 - x^2)}{(x^{12} + 3x^6 + 1) \tan^{-1} \left(x^3 + \frac{1}{x^3} \right)} dx \\ &= \int \frac{x^6 \left(x^2 - \frac{1}{x^4} \right)}{x^6 \left(x^6 + 3 + \frac{1}{x^6} \right) \tan^{-1} \left(x^3 + \frac{1}{x^3} \right)} dx \\ &= \int \frac{\left(x^2 - \frac{1}{x^4} \right) dx}{\left(\left(x^3 + \frac{1}{x^3} \right)^2 + 1 \right) \tan^{-1} \left(x^3 + \frac{1}{x^3} \right)} \end{aligned}$$

$$\text{Put } \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) = t$$

$$\Rightarrow \frac{1}{1 + \left(x^3 + \frac{1}{x^3} \right)^2} \times 3 \left(x^2 - \frac{1}{x^4} \right) dx = dt$$

$$\Rightarrow \frac{1}{3} \int \frac{dt}{t} = \frac{1}{3} \ln \left| \tan^{-1} \left(x^3 + \frac{1}{x^3} \right) \right| + C$$

Question: $S_n = 2023\alpha^n + 2024\beta^n$, $x^2 - x - 1 = 0$

Options:

(a) $2S_{12} = S_{11} + S_{10}$

(b) $S_{12} = S_{10} + S_{11}$

(c) $S_{12} = S_{10} + S_{11}$

(d) $2S_{12} = S_{10} + S_{11}$

Answer: (a)

Solution:

$$ax^2 + bx + c = 0$$

$$aS_n + bS_{n-1} + cS_{n-2} = 0$$

$$S_{12} - S_{11} - 1 \times S_{10} = 0$$

Question: Consider an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b)$; eccentricity e_1 & Hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, with eccentricity e_2 , $e_1 e_2 = 1$. Ellipse passes through the foci of hyperbola find the length of the ellipse along $y = 2$.

Answer: $\frac{10\sqrt{5}}{3}$

Solution:

Hyperbola

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

$$e_1 = \sqrt{1 + \frac{9}{16}} = \frac{5}{4}$$



For ellipse

$$2a = 2 \times 4e_1$$

$$a = 4e_1$$

$$a = 5$$

Also

$$e_1 e_2 = 1$$

$$\frac{5}{4} e_2 = 1$$

$$e_2 = \frac{4}{5}$$

By ellipse formula

$$b^2 = a^2 (1 - e_2^2)$$

$$b = 3$$

$$\text{Ellipse : } \frac{x^2}{25} + \frac{y^2}{9} = 1$$

Put $y = 2$ in ellipse equation

$$x_1^2 = \frac{25 \times 5}{9}$$

$$x_1 = \frac{5\sqrt{5}}{3}$$

Length of the chord

$$\Rightarrow l = 2x_1 = \frac{10\sqrt{5}}{3}$$

Question: If $2 \tan^2 \theta - 5 \sec \theta = 1$ has exactly 7 solutions in $\left[0, \frac{n\pi}{2}\right]$ least value of $n \in N$,

then $\sum_{k=1}^n \frac{k}{2^n}$ is equal to _____.

Options:

(a) $\frac{9}{2^9}$

(b) $\frac{91}{2^{13}}$

(c) $\frac{7}{2^7}$

(d) $\frac{11}{2^{12}}$

Answer: (b)

Solution:

$$2 \tan^2 \theta - 5 \sec \theta = 1$$

$$2(\sec^2 \theta - 1) - 5 \sec \theta = 1$$

$$2 \sec^2 \theta - 5 \sec \theta - 3 = 0$$

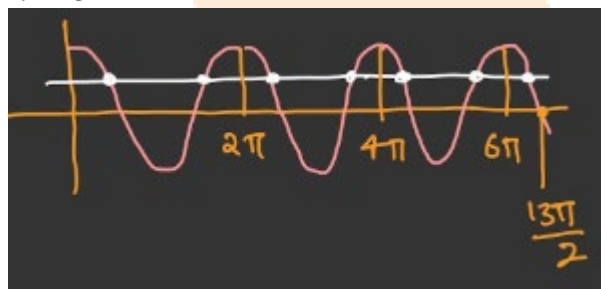
$$\Rightarrow (2 \sec \theta + 1)(\sec \theta - 3) = 0$$

$$\Rightarrow \sec \theta = \frac{-1}{2}, \sec \theta = 3$$

$$\cos \theta = -2, \cos \theta = \frac{1}{3}$$

$$7 \text{ solutions in } \left[0, \frac{n\pi}{2}\right]$$

$$n = 13$$



$$\sum_{k=1}^{13} \frac{k}{2^{13}} = \frac{1}{2^{13}} (1 + 2 + \dots + 13)$$

$$= \frac{1}{2^{13}} \times \left(\frac{13 \times 14}{2} \right) = \frac{91}{2^{13}}$$

Question: $\lim_{x \rightarrow 0} \frac{3 - a \sin x - b \cos x - \log(1+x)}{3 \tan^2 x}$ is non zero finite. Find $2b - a = ?$

Answer: 7.00

Solution:

$$\lim_{x \rightarrow 0} \left(\frac{3 - a \sin x - b \cos x - \ln(1+x)}{3 \tan^2 x} \right)$$

$$\lim_{x \rightarrow 0} \frac{\left(3 - a \left(x - \frac{x^3}{6} \right) - b \left(1 - \frac{x^2}{2} \right) - \left(x - \frac{x^2}{2} \right) \right)}{3x^2}$$

Coeff. of x^0 : $3 - b = 0 \Rightarrow b = 3$

Coeff. of x : $-a - 1 = 0 \Rightarrow a = -1$

$\Rightarrow 2b - a = 7$

Question: $\int_0^{\pi} \frac{dx}{1 - 2a \cos x + a^2}$ is equal to:

Options:

(a) $\frac{(1+a^2)\pi}{(1-a^2)^2}$

(b) $\frac{\pi}{(1-a^2)}$

(c) $\frac{(1-a^2)\pi}{(1+a^2)}$

(d) $\frac{(1-a^2)\pi}{(1+a^2)^2}$

Answer: (b)

Solution:

$$I = \int_0^{\pi} \frac{dx}{1 - 2a \cos x + a^2} \quad I = \int_0^{\pi} \frac{dx}{1 + 2a \cos x + a^2}$$

$$2I = \int_0^{\pi} \frac{2(1+a^2)}{(1+a^2)^2 - 4a^2 \cos^2 x}$$

$$2I = 2 \int_0^{\frac{\pi}{2}} \frac{2(1+a^2)}{(1+a^2)^2 - 4a^2 \cos^2 x}$$

$$I = \int_0^{\frac{\pi}{2}} \frac{2(1+a^2) \sec^2 x}{(1+a^2)^2 (1 + \tan^2 x) - 4a^2} dx$$

$$I = \int_0^{\frac{\pi}{2}} \frac{\left(\frac{2}{1+a^2} \right) \sec^2 x}{\tan^2 x + \left(\frac{1-a^2}{1+a^2} \right)} dx$$

Put $\tan x = t$

$$\begin{aligned}
 I &= \frac{2}{1+a^2} \int_0^{\infty} \frac{dt}{t^2 + \left(\frac{1-a^2}{1+a^2}\right)^2} \\
 &= \left(\frac{2}{1+a^2}\right) \times \left(\frac{1+a^2}{1-a^2}\right) \times \left| \tan^{-1} t \frac{(1+a^2)}{(1-a^2)} \right|_0^{\infty} \\
 &= \frac{2}{1-a^2} \times \frac{\pi}{2} \\
 &= \frac{\pi}{1-a^2}
 \end{aligned}$$

Question: If the area $\{(x, y) : 0 \leq y \leq \min(2x, 6x - x^2)\}$ is A then 12 A ____.

Answer: 304.00

Solution:

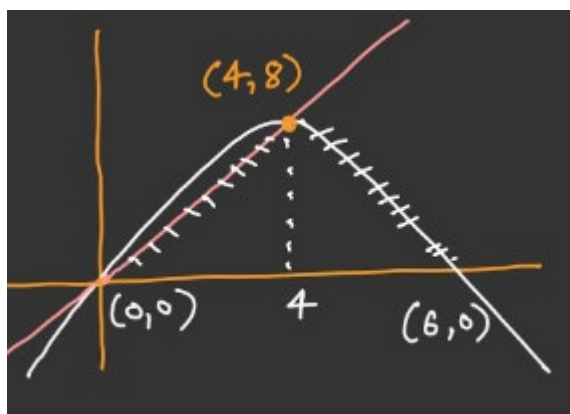
Given, $0 \leq y \leq \min(2x, 6x - x^2)$

$$6x - x^2 = 2x$$

$$x^2 = 4x$$

$$x = 0, x = 4$$

$$A = 16 + \int_4^6 (6x - x^2) dx$$



$$A = 16 + \left[3x^2 - \frac{x^3}{3} \right]_4^6$$

$$12A = 304$$

Question: If mean of 15 observation is 12 & s.d. = 3. 10 was taken wrongly in place of 12.

New mean = μ , Variance = σ^2 . Find the value = $15(\mu + \mu^2 + \sigma^2)$

Answer: 2521.00

Solution:

$$\frac{(x_1 + x_2 + \dots + x_{14}) + 10}{15} = 12 \quad \dots(1)$$

$$\frac{(x_1 + x_2 + \dots + x_{14}) + 12}{15} = \mu \quad \dots(2)$$

$$(2) - (1)$$

$$\frac{2}{15} = \mu - 12$$

$$\mu = \frac{182}{15}$$

$$\frac{x_1^2 + \dots + x_{14}^2 + 10^2}{15} - 12^2 = 9 \quad \dots(3)$$

$$\frac{x_1^2 + \dots + x_{14}^2 + 12^2}{15} - \left(\frac{182}{15}\right)^2 = \sigma^2 \quad \dots(4)$$

$$(4) - (3)$$

$$\left(\frac{12^2 - 10^2}{15}\right) - \left(\left(\frac{182}{15}\right)^2 - 12^2\right) = \sigma^2 - 9$$

$$\frac{44}{15} - \left(\frac{182}{15}\right)^2 + 144 = \sigma^2 - 9$$

$$\left(\frac{182}{15}\right)^2 + \sigma^2 = \frac{44}{15} + 144 + 9$$

$$\mu^2 + \sigma^2 + \mu = \frac{44}{15} + 153 + \frac{182}{15}$$

$$15(\mu + \mu^2 + \sigma^2) = 44 + 182 + 153 \times 15 = 2521$$

Question:
$$\begin{vmatrix} 1 & \frac{3}{2} & \alpha + \frac{3}{2} \\ 1 & \frac{1}{3} & \alpha + \frac{1}{3} \\ 2\alpha + 3 & 3\alpha + 1 & 0 \end{vmatrix} = 0$$
 then α lies in

Options:

(a) $\left[-\frac{3}{2}, \frac{3}{2}\right]$

(b) $(-3, 0)$

(c)

(d)

Answer:

Solution:

$$\left(\alpha + \frac{3}{2}\right)\left[3\alpha + 1 - \frac{2\alpha}{3} - 1\right] - \left(\alpha + \frac{1}{3}\right)\left[3\alpha + 1 - 3\alpha - \frac{9}{2}\right] = 0$$

$$\left(\alpha + \frac{3}{2}\right)\frac{7\alpha}{3} + \left(\alpha + \frac{1}{3}\right)\frac{7}{2} = 0$$

$$\frac{\alpha^2}{3} + \frac{\alpha}{2} + \frac{\alpha}{2} + \frac{1}{6} = 0$$

$$2\alpha^2 + 6\alpha + 1 = 0$$

Question: Find the number of solutions: $\tan^{-1}(x) + \tan^{-1}(2x) = \frac{\pi}{4}$

Answer: 1.00

Question: A-m elements, B-n elements, subset of A is 56 more than B, P(m, n) is a point and Q(-2, -3), find distance between P and Q.

Answer: -10.00

Question: $\alpha = \frac{(4!)!}{(4!)^{3!}}$ & $\beta = \frac{(5!)!}{(5!)^{4!}}$

Options:

- (a) α is integer β is not
- (b) β is integer α is not
- (c) Both are integers
- (d) Both are not integers

Question: Let A be 2×2 real matrix and roots of equation $|A - x| = 0$ be -1 and 3. Sum of diagonal elements of A^2 ____.

Question: $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ and $f''(x) > 0$ for $x \in (0, 3)$